



WEEK 1: ASSIGNMENT 1

Question 1:

Unit protection is based on the concept of

- a) Absolute selectivity
- b) Relative selectivity
- c) Both (a) and (b)
- d) None of these

Answer: (a)

Question 2:

Find the number of 11 kV suspension insulator discs required for a three phase 220 kV power transmission line, with a safety factor of 1.5?

- a) 12
- b) 18
- c) 20
- d) 30

Answer: (b)

Solution:

Line voltage (V_L) = 220 kV

Phase voltage (V_P) = $220/\sqrt{3}$ kV

Voltage rating of each insulator disc (V_r) = 11 kV

Safety factor (S_f) = 1.5

$$\text{Required number of 11 kV insulator discs} = \frac{V_P}{V_r} \times S_f = \frac{220/\sqrt{3}}{11} \times 1.5 = 17.32 \approx 18$$

Question 3:

What is the main drawback of electromechanical and static relays?

- a) Both are very costly.
- b) They are not rugged in nature.
- c) Lack of continuous monitoring of their operational integrity.
- d) None of these

Answer: (c)

Question 4:

A thermal relay is used for protection against

- a) Overload
- b) Overvoltage
- c) Short-circuit
- d) All the above

Answer: (a)

Question 5:

Intelligent Electronic Devices (IEDs) are useful for

- a) Protection

- b) Measurement and storing
- c) Control
- d) All of these

Answer: (d)

Question 6:

The operating time of a modern digital relay is typically of the order of

- a) 0 – 10 ms
- b) 10 – 30 ms
- c) 50 – 100 ms
- d) 100 – 200 ms

Answer: (b)

Question 7:

The VA rating of a current transformer (CT) is determined by considering

- a) total burden connected to the CT secondary circuit.
- b) resistance of the relay only.
- c) reactance of the relay only.
- d) secondary current rating only.

Answer: (a)

Question 8:

The least number of faults is generally reported for

- a) transmission line
- b) cables
- c) switchgear
- d) transformers

Answer: (b)

Question 9:

The apparent power (VA) required to operate a relay is known as

- a) stability of the relay
- b) tripping time of the relay
- c) sensitivity of the relay
- d) None of these

Answer: (c)

Question 10:

A protective relay is used to

- (i) provide additional safety by limiting arcing current during circuit breaker operation.
- (ii) close its contacts when the actuating quantity reaches a predetermined value.

- a) Only (i)
- b) Only (ii)
- c) Both (i) and (ii)
- d) None of these

Answer: (b)

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